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Building a GPU you can actually finish

What is genuinely purchasable, the full fabrication and assembly sequence, and nine documented projects that prove where the line really sits.

3	14	9	31
REAL ROUTES TO A CARD	STEPS FROM SPEC TO VALIDATION	VERIFIED CASE STUDIES	SOURCES READ

Prepared for the GPU Builder teaching project
Companion document to the 22-step build guide

Prices indicative, checked October 2026
Availability changes weekly

01 The ceiling, stated honestly

FEASIBILITY

You cannot buy a modern GPU core and build a retail graphics card from a catalogue. The GPU package and its GDDR are supplied to board partners, and a dense multi-layer board with 0.25 mm-pitch BGA work is a fab-line job. Everything else — the board, the cooling, the connectors, the fabrication, the assembly service — is purchasable by an individual.

Partner-only
GPU cores and GDDR are not retail parts; expect salvage or donor-card supply

10–32
PCB layers on real designs; Valve's GDDR6 board is 10

0.25 mm
Finest BGA pitch PCBWay publishes for assembly

4 years
Spare-time effort behind the one finished hobby PCIe card

ROUTE	WHAT YOU ACTUALLY DO	REALISTIC COST	DOCUMENTED BY
A · FPGA graphics card	Buy an FPGA system-on-module with a hardened PCIe block, design a carrier PCB in KiCad, fabricate it, write the GPU in SystemVerilog plus a display driver	High hundreds to low thousands, dominated by the SoM and board revisions	FuryGpu; VGATonic at the cheap end
B · Donor-card rebuild or memory mod	Take a working card, move the GPU package and memory onto a new or clamshell PCB, modify firmware	≈ $430 parts + ≈ $140 clamshell kit, plus tools or a rework shop	VIK-on's 48 GB RTX 4090; 2080 Ti 22 GB attempts
C · Custom ASIC card	Design silicon, tape out, package, build the board	$2–3 M for a 100k-chip run before product costs	Open Graphics Project — and no consumer card shipped

> THE SENTENCE TO TEACH FIRST
>
> Capacity is not performance, and a recognised capacity is not a working card. The 2021 RTX 2080 Ti 22 GB attempt made the card report 22 GB and then produced severe artifacts, because the right memory packages could not be sourced. Validation, not enumeration, is what makes a build real.

02 Step by step, spec to validated card

14 STEPS

Steps 1–3 decide what you are building. Steps 4–8 are design. Steps 9–12 are the build. Steps 13–14 are what separate a demo from a card. Each step names what you buy for it.

PHASE A · DECIDE

01

Write the spec before you buy anything

Pin down output resolution and refresh, PCIe generation and lane count, memory capacity and bus width, power budget, board dimensions, and — most importantly — what software will run on it. FuryGpu's author spent almost four years and landed at "mid-1990s GPU class, Quake at a solid 60 fps". That is the realistic ceiling for route A.

SOURCE IT Nothing yet. The spec determines every later purchase.

02

Choose the engine

Route A: an FPGA SoM with a hardened PCIe core — not one built out of general logic. Barrie chose Xilinx's Kria Zynq UltraScale+ family for exactly that reason, plus its DSP and LUT budget. Check stock and price at Digi-Key, Mouser or Avnet; the KV260 starter kit is the usual entry point.

Route B: a same-model donor card. The GPU package, memory configuration, strap resistors and firmware are all model-specific, so "a 4090" is not interchangeable with "a 4090 Suprim".

SOURCE IT Digi-Key · Mouser · Avnet; eBay or a repair shop for donors.

03

Read the two standards you will be held to

PCI-SIG CEM defines the x1/x4/x8/x16 card-edge pinout, the motherboard footprint and the mating interface — the edge is a mechanical and electrical standard, not just copper fingers. Members access the specification library; non-members can purchase it.

JEDEC JESD250D covers GDDR6: 8–16 Gb, x16 dual-channel devices. Note that AC timings and capacitances are not fully standardised, so the memory vendor's datasheet governs, not the standard.

SOURCE IT pcisig.com/specifications · jedec.org · JESD250D

PHASE B · DESIGN

04

Design the power tree

You need the GPU rail (sub-1 V at very high current), the memory rail, and auxiliary rails, all sequenced so nothing is back-powered during bring-up. Multiphase buck controllers are the norm: study TI's Multiphase 101 series for the equations and lab method, then look at a real board — GamersNexus counted a 16-phase VRM on the Titan V.

SOURCE IT Controllers, DrMOS/power stages, inductors and capacitors from Mouser or Digi-Key; TI reference designs.

05

Schematic capture

KiCad is free and is what FuryGpu used. The sheet contains the SoM or GPU footprint, memory, power tree, PCIe edge, display connectors, fan headers, and the SPI flash that holds firmware. Draw the power sequencing and test points deliberately — you will need them at step 11.

SOURCE IT KiCad (free). Nothing to buy.

06

Talk to the fabricator before you lay out

This is the step hobbyists skip and then lose a month. Ask for the stackup, the impedance tolerance, **electrolytic hard gold** on the edge fingers (ENIG gold is not the same thing), and a bevel angle. Confirm they can do all of it on one order. FuryGpu's board went through several JLCPCB revisions, fixing issues that only appeared once a board arrived.

SOURCE IT PCBWay · Sierra Circuits · Bay Area Circuits · Eurocircuits · JLCPCB (verify hard gold).

07

Lay out the board

BGA escape, memory routing, PCIe differential pairs, solid reference planes, thermal copper. Three facts worth internalising: layer count is design-dependent, not a rule — Valve's GDDR6 GPU board is 10 layers; their engineer matches signal arrival delay rather than physical length, because propagation speed differs per layer; and they used backdrilling to remove via stubs. AMD's public fly-by versus clamshell guidance is written for DDR4, so use it to teach the concept, never as a GDDR6 recipe.

SOURCE IT Cadence's GDDR6 SI/PI paper for the validation workflow; AMD UG583 for topology concepts.

08

Design the cooling and the mechanicals

Heatsink or vapour chamber, fan curve, pad thickness over memory and VRM, backplate, PCIe bracket, M2.5 standoffs, and the airflow path. On a 4090, GamersNexus measured roughly 5 °C just from blocking the flow-through opening — airflow is part of the electrical design, not an accessory.

SOURCE IT Thermal Grizzly or ModMyMods pads and paste; Noctua or Delta fans; McMaster-Carr M2.5 hardware; exact-match brackets from eBay spares.

PHASE C · BUILD

09

Fabricate the bare board

Quote it as the product it is. A $5 two-layer prototype is not the same object as a 10-layer controlled-impedance card with hard-gold fingers, and a fabricator's advertised layer count does not by itself prove they can build your stackup. Ask for the impedance coupon and the edge-contact drawing with the quote.

SOURCE IT PCBWay (14 layers standard, 0.15 mm min annular ring, 20–60° bevels) · Sierra (HDI to 30 layers, ±5 %) · OSH Park for 4-layer only, no hard gold.

10

Assemble and inspect

Stencil, paste, place, reflow, inspect. Hidden BGA joints need X-ray — you cannot eyeball them, and a hot-air station alone is not a GPU BGA process. A typical SAC305 profile peaks at 235–245 °C with 45–90 s above liquidus, but that is an example, not a recipe: build the profile for your paste, board and components. Underfill is optional and only where the package calls for it.

SOURCE IT Turnkey assembly from PCBWay, Eurocircuits (from one board) or Sierra; reball service at Joshi Repair (£60 GPU, £10/chip memory).

11

Bring it up in order

Check every rail against its sequencing order before fitting the expensive part. Then confirm PCIe enumeration, flash the bitstream or firmware, and only then fit the cooler. Keep a dead short and a current-limited bench supply in the loop for the first power-on.

SOURCE IT Bench supply and a thermal camera or IR thermometer; SPI flash programmer if not in-system.

12

Display output and drivers

DisplayPort and HDMI mean EDID, mode setting and a real display driver. Barrie is blunt that Windows kernel driver work was the most painful part of the entire project — harder than the RTL or the board. Budget for it in the spec, not at the end.

SOURCE IT Digi-Key or Mouser board-mount HDMI/DisplayPort receptacles; the vendor's display-controller documentation.

PHASE D · PROVE

13

Assemble the cooler and check fit

Torque the cooler evenly, verify pad contact on memory and VRM, fit the backplate and bracket, and check the card physically seats in a real slot with a real chassis clearance. A card that only fits a test bench is not finished.

SOURCE IT Model-specific fans and heatsinks from GPU Fan Replacement or eBay; Noctua for deshroud builds.

14

Validate, then label it honestly

Memory stress, thermal soak, benchmark loops, power draw at the wall, and a long stability run. Then describe the result for what it is: display-output verified, FPGA or ASIC design, board modification, or planned. That label is the difference between a case study and a claim.

SOURCE IT Nothing to buy — this step is time and measurement.

03 Where the parts actually come from

SOURCING

Four categories behave completely differently. Ordinary electronics are catalogue parts; mechanical parts are model-specific; memory is broker supply; the GPU package has no retail channel at all.

Fabrication and assembly

SUPPLIER	WHAT THEY PUBLISH	WATCH FOR
PCBWay	1–14 layers standard, HDI and high-layer on request; 0.15 mm minimum annular ring; hard gold and 20/30/45/60° bevels; BGA assembly to 0.25 mm pitch with X-ray	Quote the stackup, gold and bevel together
Sierra Circuits	HDI to 30 layers, rigid to 36; ±5 % advanced impedance; electrolytic hard gold; 20–45° bevels	Premium quote; ask for the impedance coupon
Bay Area Circuits	Up to 30 layers, HDI, ±5 % advanced with TDR testing, hard gold, 20–60° bevels	US turnaround; custom quote
Eurocircuits	16-layer pool, defined-impedance pool at 4/6/8 layers, gold edge connectors and bevelling; assembly from a single PCB	Best small-run assembly route in the EU
JLCPCB	1–32 layers, 1–3-step HDI, ±10 % impedance, gold fingers, cheap prototypes and standard PCBA	Gold fingers are gold under ENIG — confirm hard gold explicitly; economic PCBA excludes gold fingers
OSH Park	4 layers, fixed stackup, $10/in² for three boards	No hard gold, no bevel — not a card edge

Memory, cores and rework

ITEM	WHERE IT COMES FROM	INDICATIVE PRICE
GDDR6 / GDDR6X	Micron catalogue and sample route; brokers, AliExpress, eBay; harvested from dead cards	Joshi Repair: 1 GB £15, 2 GB £20, GDDR6X 1 GB £25, 2 GB £40
GPU package	Same-model donor card, or a repairer who sources it. Salvage cores online are unverified provenance	Donor-card dependent
Reballing service	Joshi Repair; Circuits Central with pre- and post-work X-ray	£60 GPU, £10 per memory chip; X-ray quoted
Hot air	Aoyue 852A++	$149.99

ITEM	WHERE IT COMES FROM	INDICATIVE PRICE
Reflow oven	VEVOR T-962A	$266.99
Integrated BGA station	AOYUE BGA9000A	≈ $10,137
X-ray	Refurbished systems start under $25,000 — use a service instead	Job quoted

Everything around the board

PART	SOURCE	NOTE
Fans, heatsink, backplate, bracket	GPU Fan Replacement, eBay spares, Noctua, Delta/Sunon via Digi-Key	Model-specific; measure before ordering
Thermal pads and paste	Thermal Grizzly, ModMyMods	Kryonaut 1 g $11.58; pad thickness is design-specific
8-pin / 12V-2×6 power	Molex Mini-Fit via Digi-Key/Mouser; MODDIY for 12VHPWR	12VHPWR from $2.99 — verify revision and rating
HDMI / DisplayPort	Digi-Key, Mouser board-mount receptacles	Match footprint and high-speed requirements
Standoffs, screws	McMaster-Carr M2.5, electronics assortments	Ordinary catalogue parts
VRM silicon	Mouser, Digi-Key, TI reference designs	Generic parts are easy; the validated layout is not

> **THE REALISTIC PURCHASING PLAN**
>
> Do not try to buy a new GPU core from a distributor. Buy a donor card for the GPU and its memory, commission a bare board with hard gold and a bevel, and pay a rework shop for the BGA work and X-ray. Everything else — cooling, hardware, connectors, paste — is ordinary retail.

04 Case studies

VERIFIED

Each entry was checked against a primary page — the project's own writing, the chip page, or a hands-on report. Cost figures are the ones the sources state.

FuryGpu

DISPLAY VERIFIED

2021–2024 · Dylan Barrie

BUILT	A real PCIe graphics card: Kria Zynq UltraScale+ SoM on a custom KiCad board fabricated at JLCPCB, DisplayPort and HDMI outputs, 4-lane PCIe, custom graphics API, and **Windows kernel display and audio drivers**.
NUMBERS	Almost four years of spare time, several board revisions, Quake at a solid 60 fps. Capability comparable to a high-end mid-1990s GPU.
LESSON	The template for route A. Board design was hard; the **driver** was the hardest part. Start small: he began with an Arty Z7 and a spinning cube.

furygpu.com blog/hello tweaktown

TinyGPU · TT07

ASIC DESIGN

2024 · Tiny Tapeout

BUILT	A GPU **taped out as an ASIC** on a Tiny Tapeout shuttle, driving a VGA monitor through a TinyVGA PMOD. Transformation, lighting, rasterisation and affine texture mapping; geometry fed from a PC over serial.
NUMBERS	50 MHz, 640×480 at 6-bit RGB, two textured triangles, no framebuffer or line buffer.
LESSON	The cheapest real silicon path that exists — and a precise demonstration of why scale limits are architectural, not just budget.

tinytapeout.com · tt07 github

TinyGPU v2

ASIC DESIGN

SKY25b shuttle

BUILT	A standalone version reading models from QSPI flash with gamepad input, 320×240 at 4-bit colour, z-buffered, up to 1,000 triangles.
NUMBERS	6.5 fps at 25 MHz; a documented viewport-edge condition could freeze the GPU until reset, fixed in v3.
LESSON	Rare public evidence of **iteration between silicon revisions** — the honest version of a roadmap.

tinytapeout.com · v2 github

VGATonic

DISPLAY VERIFIED

circa 2014 · PK

BUILT	A 144-macrocell CPLD video adapter with a microcontroller, home-made PCB and SPI interface, plus Raspberry Pi, Linux and Arduino drivers.
NUMBERS	≈ **$24 in parts**; 640×480 at 60 Hz in RGB332; 25 fps over SPI; a Doom demo running from a Raspberry Pi 2.
LESSON	A display-producing "GPU" is achievable on a hobby budget — proof that the entry barrier is knowledge, not money.

dqydj.com · build write-up

RTX 4090 · 24 GB → 48 GB

BOARD MODIFICATION

Sept 2025 · VIK-on

BUILT	An MSI Suprim donor; the AD102 package and twelve 2 GB GDDR6X chips moved onto a **custom clamshell PCB** with memory on both sides, a blower cooler, and modified firmware so the system accepts the larger capacity.
NUMBERS	≈ $140 for the clamshell kit, ≈ $430 in parts for a DIY upgrade, before shipping.
LESSON	The closest real analogue to a banked memory layout: capacity comes from **more packages plus firmware support**, not a faster GPU. Also the clearest warning about warranty, reliability and resale.

techpowerup hothardware

RTX 2080 Ti · 11 GB → 22 GB

BOARD MODIFICATION

2021–2024

BUILT	Replacing 1 GB GDDR6 modules with 2 GB Samsung parts on an Asus 2080 Ti, with strap and firmware changes; a 2024 video documents the 11-to-22 GB procedure.
NUMBERS	Double capacity, unchanged compute. The 2021 attempt did not reach a usable card.
LESSON	The canonical "**capacity recognised ≠ working card**" case: it reported 22 GB, then showed severe artifacts, because the larger packages could not be sourced.

videocardz upgrade video

3D Graphics Thingy

ABANDONED

pre-2013 · Big Mess o' Wires

ATTEMPTED	An ambitious FPGA 3D graphics processor that never reached its graphics features.
NUMBERS	Died of DDR2 bandwidth, a memory-controller side quest, and fighting Xilinx tooling.
LESSON	The author's own conclusion: start small and expand incrementally. The successor plan used the DE1 board's onboard SRAM instead of chasing external memory.

bigmessowires.com · post-mortem

OGD1 · Open Graphics Project STOPPED SHORT 2006–2008

BUILT	A real open-hardware PCI/PCI-X graphics board: two FPGAs, 256 MB RAM, DVI and S-video outputs, schematics released.
NUMBERS	**$1,500** preorder, $1,400 for the first hundred. A 100k-chip ASIC run was estimated at **$2–3 M** before other product costs.
LESSON	The board shipped as a development platform, not a working GPU, and the consumer ASIC card never arrived. The clearest cost-reality case study in the field.

engadget ASIC economics interview

Boundary cases — label these honestly

Vortex
github.com/vortexgpgpuRISC-V GPGPU, 32-core Stratix 10 configuration, PCIe and OpenCL. No verified direct monitor output — an acceleration and architecture study.

Nyuzi
github.com/jbush001Experimental synthesizable GPGPU with graphics programs, but compute-focused by its own documentation. No verified display card.

MIAOW
miaowgpu.orgAn open RTL implementation of an AMD compute unit; its authors state it lacks graphical output and a memory interface.

FIFOgfx
github.com/MagerValpOpen graphics adapter for 8/16-bit machines, under-$50 BOM, documented design rather than a verified finished build.

What actually goes wrong

Sourcing beats skill. Every failed or stalled board in this research failed at a part, not at the soldering: unavailable 2 GB GDDR6 modules, no retail GPU core, a 100k-chip run that needed millions.

The driver is the long pole. FuryGpu's author calls Windows driver work the most painful part of four years — more than the RTL or the PCB. A card with no driver is a bench ornament.

Scope creep kills FPGA projects. 3DGT died chasing external memory bandwidth and tooling. The projects that shipped started with one triangle or one spinning cube.

Validation is the product. The 2080 Ti mod reported the right capacity and produced artifacts. Reporting and working are different states.

Cheapness is legitimate. VGATonic cost $24 and produced a real picture. It is a better first project than a 10-layer PCIe card.

Memory on both sides is a real technique. The 4090 clamshell mod doubles capacity by doubling package count — the same principle a banked build teaches.

How to present these on the site

Give every case one of four labels, and never remove it: **display-output verified**, **FPGA or ASIC design**, **board modification**, **planned or abandoned**. That single field stops an unprogrammed OGD1 or a MIAOW compute unit from reading as a finished graphics card, and it gives the failure cases as much standing as the successes.

Where this connects to the current build guide

- The guide's sourcing classes map directly onto reality: hardware and distributor parts are genuinely buyable, fabrication is commissionable, and integrated parts (GPU package, GDDR) have no retail channel — which is exactly what the donor-card route is for.
- The custom builder's bank model has a documented precedent: the 48 GB clamshell conversion adds capacity by adding packages on both sides of the board and changing firmware.
- Any step that says "solder the memory" should say who solders it, at what pitch, and with what inspection — otherwise the lesson teaches a process nobody can safely finish at home.

> CAVEATS WORTH PRINTING ON THE PAGE
>
> Prices were checked in October 2026 and move weekly, especially on memory. Fabricator capability tables describe what a vendor advertises; confirm your specific stackup, hard gold and bevel in writing. Reflow ovens and hot-air stations are mains-powered, reach 250 °C and produce lead-free fumes — extraction and fire safety are part of the lesson, not a footnote. Nothing here is a promise that a home-built card will be stable, warrantable or resellable.

Sources

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Building a GPU you can actually finish · research brief prepared for GPU Builder, October 2026. Companion to the interactive teaching guide and the custom memory-bank builder. Every figure in this document is attributed to a source above; where a source stated a limitation, the limitation is reproduced rather than smoothed over.